

Chapter 6: Functions

The Building Blocks of Trading Systems

On a chilly January morning in 2012, Ray Dalio, founder of Bridgewater Associates—the world's largest hedge fund managing over \$160 billion—made a remarkable announcement. His firm would begin codifying its investment principles into algorithms to make better trading decisions than humans could on their own. This approach, which Dalio called "systematic decision-making," would ultimately transform Bridgewater into one of the most technologically sophisticated investment firms in the world.

At the heart of Bridgewater's transformation were functions—reusable blocks of code that perform specific tasks. These digital building blocks allowed the firm to transform abstract investment principles into concrete, repeatable operations that could be tested,